c. Applications

1. It is used in homes
2. It is used in hotels and restaurants
3. It is used in shops
4. It is used in offices
5. It is used in roads, streets and parks
6. It is used in hospitals
7. It is used in commercial premises

2.10 Lighting schemes

Lighting schemes are classified according into their locations requirement and purposes etc., as stated under:

i. Direct lighting
ii. Indirect lighting
iii. Semi direct lighting
iv. Semi indirect lighting
v. General lighting

i. Direct lighting

In this system almost 90 to 95% of light falls directly on the object or the surface. The light is made to fall upon the surface with the help of deep reflectors. Such a type of lighting scheme is mostly used in industries and commercial lighting. Although this scheme is most efficient, it is liable to cause glare and shadows.

ii. Indirect lighting

In this system, the light does not fall directly on the surface but more than 90% of light is directed upwards by using diffusing reflectors. Here the ceiling acts as a source of light and this light is uniformly distributed over the surface and glare is reduced to minimum. It provides shadow-less illumination which is useful for drawing offices and composing rooms. It is also used for decoration purposes in cinema halls, hotels, etc.

iii. Semi direct lighting

This is also an efficient system of lighting and chances of glare are also reduced. Here transparent type shades are used through which about 60% of light is directed downward and 40% is directed upward. This also provides a uniform distribution of light and is best suited for room with high ceilings.

iv. Semi indirect lighting

In this system about 60 to 90% of total light is thrown upward to the ceiling for diffused reflection and the rest reaches the working plane directly. A very small amount of light is absorbed by the bowl. It is mainly used for interior decoration.

v. General lighting

This system employs such type of luminaries, shades and reflectors which give equal illumination in all the directions.

2.10.1 Design of indoor light scheme

While designing a good lighting scheme, the following points must be kept in mind:

i. It should provide adequate illumination.
ii. It should provide uniformly distributed light all over working plane.
iii. It should avoid glare and shadows as far as possible.
iv. It should provide light of suitable colours.

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